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API-driven Cloud Native Solutions

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CC ZG506 / SE ZG506

Lecture No. 10

Agenda

Part-B: Language Models

- Open AI's API
- Fine Tuning a Model using Open AI's API
- Differences between “Fine Tuning” and “Context based Approach” for document Q&A
- Visual Models for Large Language Modeling Experiments (Rivet)



Open AI API

Open AI API

- OpenAI offers several APIs that allow developers to integrate powerful AI capabilities into their applications.
- There are Five types of APIs offered by Open AI
 1. Chat API / Completion API
 2. Embedding API
 3. DALL-E API
 4. Whisper API
 5. Fine-tuning API



Open AI API

Before starting with any of the APIs, first we need to create the “Access Token”

URL - <https://platform.openai.com/docs/overview>

Create Access Token at -> <https://platform.openai.com/api-keys>

Under “Dashboard” -> “API Keys”

Check Usage at -> <https://platform.openai.com/usage>

Under “Dashboard” -> “Usage”

Open AI API

Base Code in “Google Colab” or VS Code

```
# install openai
```

```
pip install openai
```

```
# provide the API Key
```

```
import openai
```

```
openai.api_key = "your_api_key"
```



1. Chat API / Completion API

- Used to create conversational AI with natural language understanding and generation capabilities.
- The API powers chatbot applications, customer service bots, and personal assistants.
- It supports multi-turn conversations and can remember context within sessions.
- Popular models include "gpt-3.5-turbo", "gpt-4", "gpt-4o-mini"

2. Embedding API

- Creates high-dimensional vector representations (embeddings) of text, which can be used for tasks like similarity matching, classification, and clustering.
- Used for information retrieval, sentiment analysis, recommendation engines, and similarity-based search functionalities.

3. DALL-E API

- Generates images from textual descriptions.
- Used to create images for various applications, including creative content generation, advertising, and prototyping.

4. Whisper API

- Transcribes spoken language into text.
- Provides **speech-to-text translation with multi-language support** and is highly accurate for clear audio sources.

DEMO

- Google Colab link ->
- <https://colab.research.google.com/drive/19YqKuCn5fUUHMEhDSUe2qnoxjLX4u5lC?usp=sharing>

5. Fine-tuning API

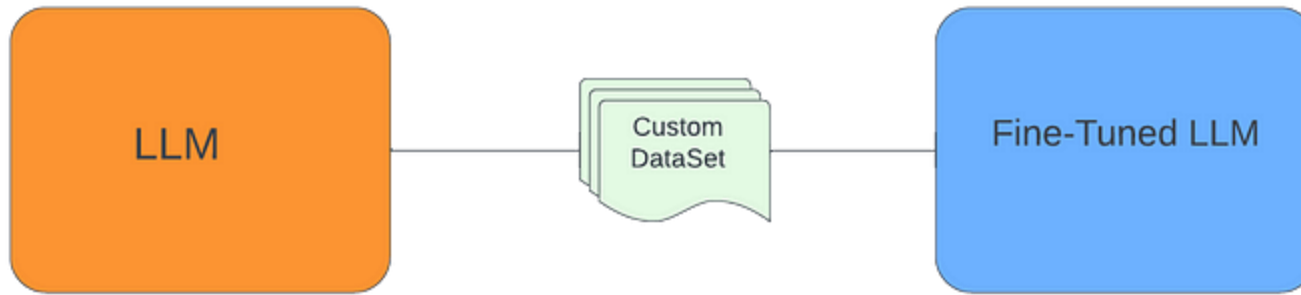
- Allows developers to fine-tune models (e.g., GPT-4o-mini) with custom datasets to perform specialized tasks more effectively.
- Ex: Customer service bots, process documents, tailored educational models, industry-specific AI applications.

Pre-requisite

- Before Fine-Tuning, we have to prepare our sample dataset in **JSONL format**

Ex: `{"messages": [{"role": "system", "content": "You are a helpful assistant."}, {"role": "user", "content": "Is the dissertation project a group project or an individual project?"}, {"role": "assistant", "content": "It has to be an individual project. Group projects are not allowed."}]}`

What is Fine-Tuning a LLM



- Fine-tuning LLM involves the additional training of a pre-existing model, which has previously acquired patterns and features from an extensive dataset, using a smaller, domain-specific dataset.

Ref: <https://dassum.medium.com/fine-tune-large-language-model-llm-on-a-custom-dataset-with-qlora-fb60abdeba07>

What is Fine-Tuning a LLM

Steps

1. Select a pre-trained LLM model [Ex: “gpt-4o-mini” from Open AI]
2. Gather relevant Dataset [Should be in the “JSONL” format]
3. Pre-process the dataset for any errors or inconsistencies
4. Fine-Tune [Fine tune the pre-trained LLM model for the specific dataset, and create a “New Model” in the process]
5. Task-specific adaptation [process retains the general language knowledge gained during pre-training while tailoring the model to the nuances of the target domain]

Ref: <https://dassum.medium.com/fine-tune-large-language-model-llm-on-a-custom-dataset-with-qlora-fb60abdeba07>

OpenAI's models for Fine Tuning

- gpt-4o-2024-08-06
- gpt-4o-mini-2024-07-18
- gpt-4-0613
- gpt-3.5-turbo-0125
- gpt-3.5-turbo-1106
- gpt-3.5-turbo-0613

<https://platform.openai.com/docs/guides/fine-tuning>

Differences

Aspect	"Context" based Approach or Prompt Engineering	Fine-Tuned Model
Context Handling	Context provided manually for each query [Word or PDF document] No model re-training	No external context needed; model is pre-trained
Knowledge Source	Limited to the document uploaded	Learns company-specific knowledge from fine-tuning
Scalability	Limited to the uploaded document	Best for focused tasks related to fine-tuned domain
Customization	Uses a pre-trained, general-purpose model	Tailored to company data through additional training
Domain Specialization	General-purpose, non-specialized	Highly specialized through fine-tuning



Demo of Fine Tuning

Exercise (Self Study)

- Explore other SLMs or LLMs like Llama 3 (Open Source) or DeepSeek and use their APIs to carry out the fine-tuning.



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Rivet

Rivet

- Rivet is an open source visual programming environment for building AI agents with LLMs.
- Iterate on prompt graphs in Rivet, then run them directly in your application.
- With Rivet, teams can effectively design, debug, and collaborate on complex LLM prompt graphs, and deploy them in their own environment.

Rivet Tool Demo



Thank You!